

Quantitative Architectural Analysis: A New Approach to Cortical Mapping

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Abstract Results from functional imaging studies are often still interpreted using the classical architectonic brain maps of Brodmann and his successors. One obvious weakness in traditional, architectural mapping is the subjective nature of localizing borders between cortical areas by means of a purely visual, microscopical examination of histological specimens. To overcome this limitation, objective mapping procedures based on quantitative cytoarchitecture have been generated. As a result, new maps for various species including man were established. In our contribution, principles of quantitative cytoarchitecture and algorithm-based cortical mapping are described for a cytoarchitectural parcellation of the human auditory cortex. Defining cortical borders based on quantified changes in cortical lamination is the decisive step towards a novel, highly improved probabilistic brain atlas.

Keywords Cytoarchitecture · Gray level index · Cortical area · Border detection · Brain mapping

Over the last 20 years, considerable advances have been achieved in the field of functional brain imaging. Using cognitive paradigms, significant brain activations can now be detected at distinct sites in the human neocortex. However, for the interpretation of such activation clusters, topographical maps of the human cortex are still in use which were established about a century ago. Prominent examples are the cytoarchitectonic map of Brodmann (1909) and its adaptation in the atlas of Talairach and Tournoux (1988). Brodmann subdivided the cortex into approximately 50 cytoarchitectonically distinct areas, and summarized his findings in a schematic drawing of a ‘typical’ brain.

Trying to relate functional findings to such a map reveals severe drawbacks, e.g., the maps are two-dimensional. Therefore, they can’t be superimposed to 3D brain-volumes as generated in MRI-studies. Brodmann’s map shows areal borders defined in only one hemisphere, thus ignoring the interindividual and interhemispheric variations in areal location, size and shape. Moreover, two-thirds of the cortical surface remain hidden in the depth of the sulci (Zilles et al. 1989) and hence do not appear in the schematic drawing of a surface of a brain. Finally, maps are biased by subjective criteria since only visual, microscopical examination of the histological specimens was used to establish the areal borders. In consequence, various brain atlases were published which exhibit remarkable differences in the topography of cortical areas.

Contradictory parcellation schemes of classical brain atlases are demonstrated in Fig. 1, depicting Brodmann’s map (Brodmann 1909) and Sarkisov’s map (Sarkisov et al.

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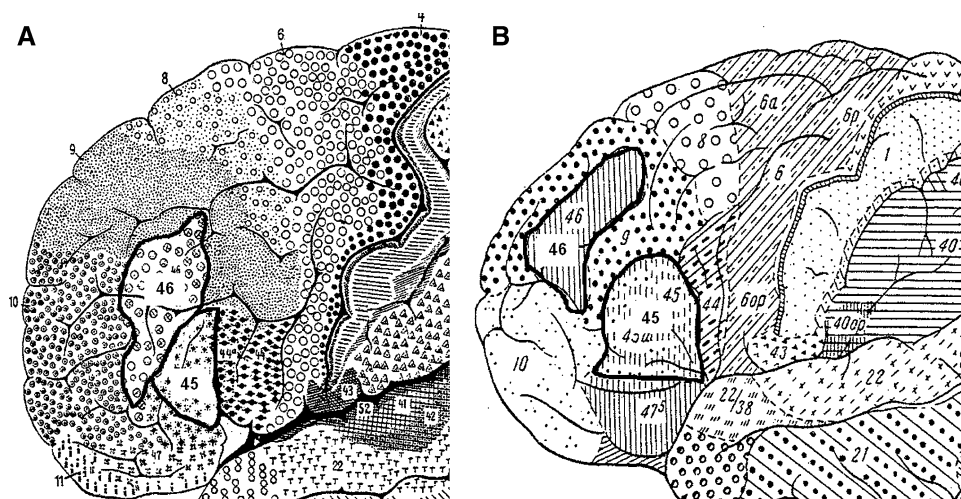
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Fig. 1 Cytoarchitectonic maps of the lateral surface of the human brain adapted from **a** Brodmann (1909) and **b** Sarkisov et al. (1949). Cytoarchitectonic areas 45 and 46 are delineated with black contours to emphasize differences in topography between the maps. Areas in both maps are classified by numerals according to Brodmann's nomenclature



1949). Substantial differences exist, e.g. with respect to the topography of area 45, a part of Broca's region, and frontal area 46. Size, position and extent differ, but most strikingly, the common border between area 45 and 46 in Brodmann's map (BA 45 and BA 46) is missing in Sarkisov's map.

An example of the enormous interindividual variability in the topography of cortical areas was demonstrated in a study on Broca's region, comprising BA 44 and 45 (Amunts et al. 2004). In addition, this study showed that macroscopical landmarks such as the sulcal pattern can not be used to establish the positions of areal borders. In all examined postmortem brains, the precentral sulcus marked approximately the caudal extent for both areas, but the precise position of the borders with respect to this sulcus varied by 1–2 cm. This can be generalized and seems to be valid for most of the secondary and multimodal associations cortices with a loose relationship between areal borders and the surrounding sulcal pattern, but also for some borders of primary areas. For example, the central sulcus always marks the posterior border of the primary motor cortex, but its anterior border is not marked by any sulcus. For a more detailed discussion and additional examples see Amunts et al. (2007). This lack of correlation between the pattern of areal borders and macroscopical landmarks excludes the possibility to transfer a location found in one brain to an other one using exclusively geometrical considerations such as the sulcal pattern. Such has been done in the atlas of Talairach and Tournoux (1988). In consequence, the areal pattern must be defined in each brain individually.

To overcome the subjective, observer-dependent nature of cytoarchitectonic brain mapping and to access the individual variability in the areal pattern, objective procedures based on quantitative cytoarchitecture have been introduced (Schleicher et al. 1999, 2005) which enable, for

the first time, a reproducible and consistent mapping of layered isocortex. This was the decisive step towards the foundation of a probabilistic atlas of the human brain (Roland and Zilles 1994; Amunts and Zilles 2006), which accounts for interindividual variability in brain morphology. In this atlas, the variability of the areal pattern influenced by methodological imperfections was dramatically reduced by replacing the traditional, subjective localization of areal borders by quantitative and, thus, more objective procedures.

Methods and Results

Laminar Pattern

The architectonic mapping of the cerebral cortex is based on the assumption of a structural–functional association of cortical areas or architectonic fields. The architecture can be established and visualized by various imaging techniques (e.g. Amunts et al. 2002; Kruggel et al. 2003; Walters et al. 2007). An architectural area is characterized by its homogeneous laminar pattern, which is different from the laminar pattern of the abutting ones. This distinction in laminar pattern reflects a functional–structural correlation, caused, for example, by differences in the input–output connectivity between areas. A key feature of areal border localization is the transition in the laminar pattern at the border between areas. A well known example is the border between the primary visual cortex BA 17 and the secondary visual cortex BA 18, which is easily recognizable in various image modalities such as cytoarchitecture, myeloarchitecture or chemoarchitecture as revealed by receptor autoradiography (Zilles et al. 2002b). This border is, among other features, characterized by an abrupt change in lamina IV, which is broad and subdivided into

sublaminae in BA 17 (IVb contains the Gennari stripe) and a thinner and less well differentiated lamina IV in BA 18.

A major problem when quantitatively analyzing the cytoarchitecture of cortical areas is the localization of the borders of a given area to the abutting ones. In order to increase the statistical power of a morphometrical study focused on distinct cortical areas, i.e. the probability of detecting existing differences between two brain samples (e.g. pathological brains versus a control sample), the variability in the data introduced by imperfections in localizing areal borders must be minimized. Our observer independent approach, which is based on multivariate, statistical tools, enables a reliable delineation of a cortical area. It is demonstrated using the auditory cortex as an example (Morosan et al. 2001), located on the superior temporal gyrus (Fig. 2).

Gray Level Index (GLI)

Transitions in the laminar pattern are objectively detected and validated in several steps. First, the shape of a laminar pattern in a cortical area, i.e. the characteristic sequence of more or less cell dense layers is quantified. The most widely used approach to shape quantification are intensity profiles (Hopf 1968a, 1968b; Schleicher et al. 1999; Schmitt and Böhme 2002). In another study, Hudspeth et al. (1976) applied microdensitometry and intensity profiles to analyze the macaque primary visual cortex using the optical density (OD) as a correlate of cell density. The intensity profiles quantified the course of the OD from the pial surface to the white matter perpendicularly to the

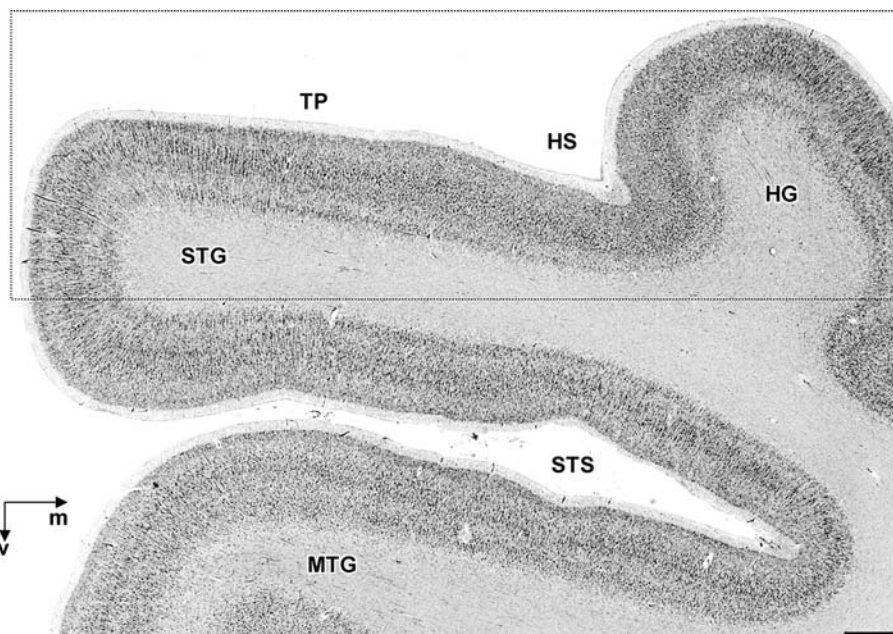
cortical layers. Hudspeth named a representative intensity profile the ‘fingerprint’ of a cortical area.

A major shortcoming of OD is its lack of stereological foundation. It is a highly biased estimate of cell density, since it depends on the staining intensity of cells and of background as well as on section thickness. Stereology is a set of mathematical procedures used to estimate structural properties of biological tissues from measurements in 2D images sampled from histological sections (Howard and Reed 1998). Since cortical tissue is highly anisotropic (pyramidal cells, dendritic bundles, minicolumns etc. exhibit a pronounced preferential orientation with regard to the cortical surface), parameters from design based stereology (Gundersen et al. 1988) such as cell size, shape or mean distance between cells cannot be applied to the analysis of serial sections with the orientation of the plane of sectioning with respect to the cortical surface changing across sections. We therefore replaced OD with a basic, but robust stereological parameter, the volume density V_V .

The visual impression of alternating cellular densities across cortical layers is mainly caused by changes in *volume density* of neurons, i.e. the *volume fraction* of stained neuronal somata per unit of volume of the reference space (the volume of a cortical area). This stereological parameter was introduced by Haug (1956), who extended the previous concept of von Economo and Koskinas (1925). He applied manual point counting to estimate the average volume density of neurons, the *gray cell coefficient*, in numerous cortical and subcortical regions in normal and pathological brains (Haug 1980, 1981).

Adhami (1973) was the first who used an image analyzing system to estimate the global volume density of

Fig. 2 Histological section in the coronal plane through the superior (STG) and middle (MTG) temporal gyri of a human brain stained for cell bodies. HG Heschl’s gyrus, HS Heschl’s sulcus, TP temporal plane, STS superior temporal sulcus. The *dotted frame* indicates the cortical ribbon used to demonstrate the sliding window procedure depicted in Fig. 8. Orientations: *m* medial and *v* ventral. Scale bar 1 mm



neurons. *Global* indicates the average of a parameter in a reference space (e.g. the volume or space of a cortical area), whereas *local* refers to values in spatially located subunits of the reference space, mostly a voxel in 3D or a cube of suitable size such as $100\ \mu\text{m}^3$. In the following years, this approach was systematically developed and applied in larger samples of histological sections (Schleicher et al. 1986; Schleicher and Zilles 1990; for details see Schleicher et al. 2000). To distinguish this automatic, local estimate of volume density from the Haug's *gray cell coefficient*, we named it the *gray level index (GLI)*.

The *GLI* is estimated as an areal density A_A , i.e. the proportion of darkly stained cellular transects to the area of the a measuring field (Fig. 3) using gray value thresholding to create a binary image from histological specimens stained for cell bodies. We analyzed histological sections stained for cell bodies using a silver stain (Merker 1983). We did not apply the classical Nissl stain (based on cresyl-violet), because the silver stain provides more robust staining results, in particular in brains which were fixed for a long time in formalin. For details of our staining protocol see Palomero-Gallagher et al. (2008). Robustness is a necessary prerequisite for further quantitative analysis. Unavoidable, subtle inhomogeneities in staining intensity were subsequently eliminated by applying an adaptive thresholding to create the binary image.

Earlier publications have demonstrated a differential shrinkage between neuropil and cell bodies in paraffin sections which is less than 10% (Kretschmann et al. 1982). We did not use celloidin embedding because we process and analyze whole brain sections of complete human brains. Although it is possible to handle whole brain sections, the quality of 3D-reconstructions from celloidin sections is lower than that of paraffin-embedded ones due to particular handling steps (e.g., free floating processing). Furthermore, freezing of adult, human brains is not feasible since, again, we are interested in processing whole brains (not small tissue blocks) in order to create a 3D atlas.

According to a basic stereological principle (Weibel 1981), A_A is an unbiased estimator of the volume density V_V , provided it is measured in planes or very thin sections. It is not dependent on the degree of anisotropy, i.e., it can be measured in images with arbitrary orientation to the cortical surface. In practical applications, however, the V_V estimate depends on two factors. First, light microscopical *GLI* estimates from standard histological sections (e.g. $20\ \mu\text{m}$ thick) are biased due to overestimation caused by projection (i.e., elements from within the depth of focus and slightly out of focus are projected onto the image plane). Second, volume densities of neurons also include non-neuronal elements such as glia and endothelial cell bodies, which cannot be reliably excluded using image analyzing procedures. However, Wree et al. (1982) found a rather constant contribution of non-neuronal elements throughout brain regions and cortical layers. Thus, this contribution increases *GLI*-values, but does not change its relative changes across the cortical layers (see discussion).

GLI-Transformation

In the first step of the localization of areal borders, a rectangular ROI is interactively defined on a histological section. Using a computer controlled scanning stage connected to a microscope, the ROI is scanned in a mosaic-like pattern into a set of tile images (Fig. 4c) (Roesch et al. 1996; Süß et al. 2002) and a *GLI*-image is calculated. An example is shown in Fig. 4a. From each tile, a binary image is created as described in Fig. 3 and subdivided into a grid of square measuring fields 16 pixels in size resulting in an in-plane resolution of $1\ \mu\text{m}$ per pixel (Fig. 4b). In each measuring field, a *GLI*-value is calculated ranging from 0% (empty field) to 100% (completely filled with cellular transects). Measuring the *GLI*-value in each field of a tile generates a submatrix of a *GLI*-image, with pixels indicating the local *GLI*-value. These matrices are

Fig. 3 Definition of the *gray level index (GLI)*. **a** Photomicrograph of isocortical layer III with prominent pyramidal cells. **b** Binary image of **a**, generated by adaptive thresholding and smoothed by a 3×3 median filter to eliminate small particles. The areal density of *white* cellular transects is 0.19 and the *GLI*-estimate 19%. Image width $180\ \mu\text{m}$ in **a** and **b**

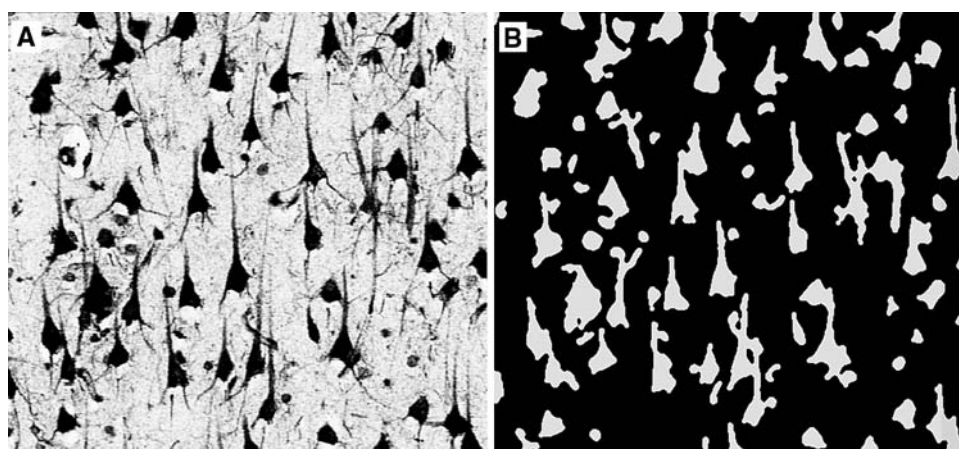
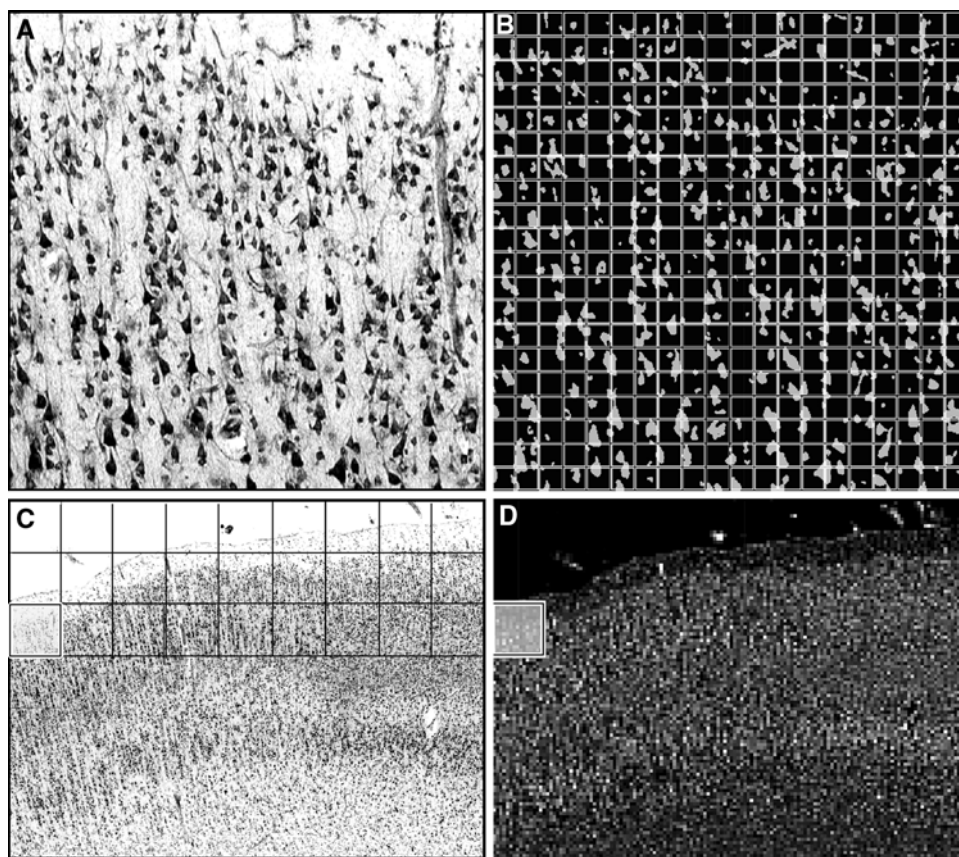


Fig. 4 *GLI*-transformation. **a** Example of a tile image acquired in the scanning procedure comprising isocortical layers I, II and III. Image width is 340 μm . **b** A grid of square measuring fields 16 μm in size is superimposed on the binary image of **a**. **c** Region of interest (ROI) scanned. The first three rows of tile images are indicated. The position of the tile image shown in **(a)** is marked by a white frame. **d** Resulting *GLI*-image. The position of the tile shown in **a** is indicated by the white frame. Each pixel codes for the *GLI* in a measuring field at a distinct position. The brighter the pixel, the higher the *GLI*-value. For details see text



combined into the resulting *GLI*-image (Fig. 4d) while scanning.

GLI-Profiles

From such *GLI*-images, *GLI*-profiles are extracted. Two examples are shown which come from a *GLI*-image of the superior temporal sulcus (Fig. 5). In a first example, a measuring line or traverse is centered in the upper left cortical sector depicted in Fig. 5a. A cortical sector is defined as a small part of the cortical ribbon delimited by two lines (traverses) running perpendicular to the cortical layers. The corresponding *GLI*-profile (Fig. 5b) quantifies the course of the *GLI* from the layer I/layer II transition (0% cortical depth) to the white matter (100% cortical depth). Details of the cytoarchitecture in this cortical sector are depicted in Fig. 6. Characteristic properties are the pronounced minicolumns with high cell density in deep layer III and a high density of small granular cells in layer IV. Both features cause distinct maxima in the *GLI*-profile (Fig. 5b). A second traverse marks a cortical sector in the upper bank of the superior temporal sulcus (Fig. 5a). To improve the visualization of the layering pattern, the two *GLI*-profiles are not individual profiles, but mean profiles averaged over 20 profiles equally spaced within each sector

(each approximately 2 mm in width). Prior to averaging, the profiles were normalized in length.

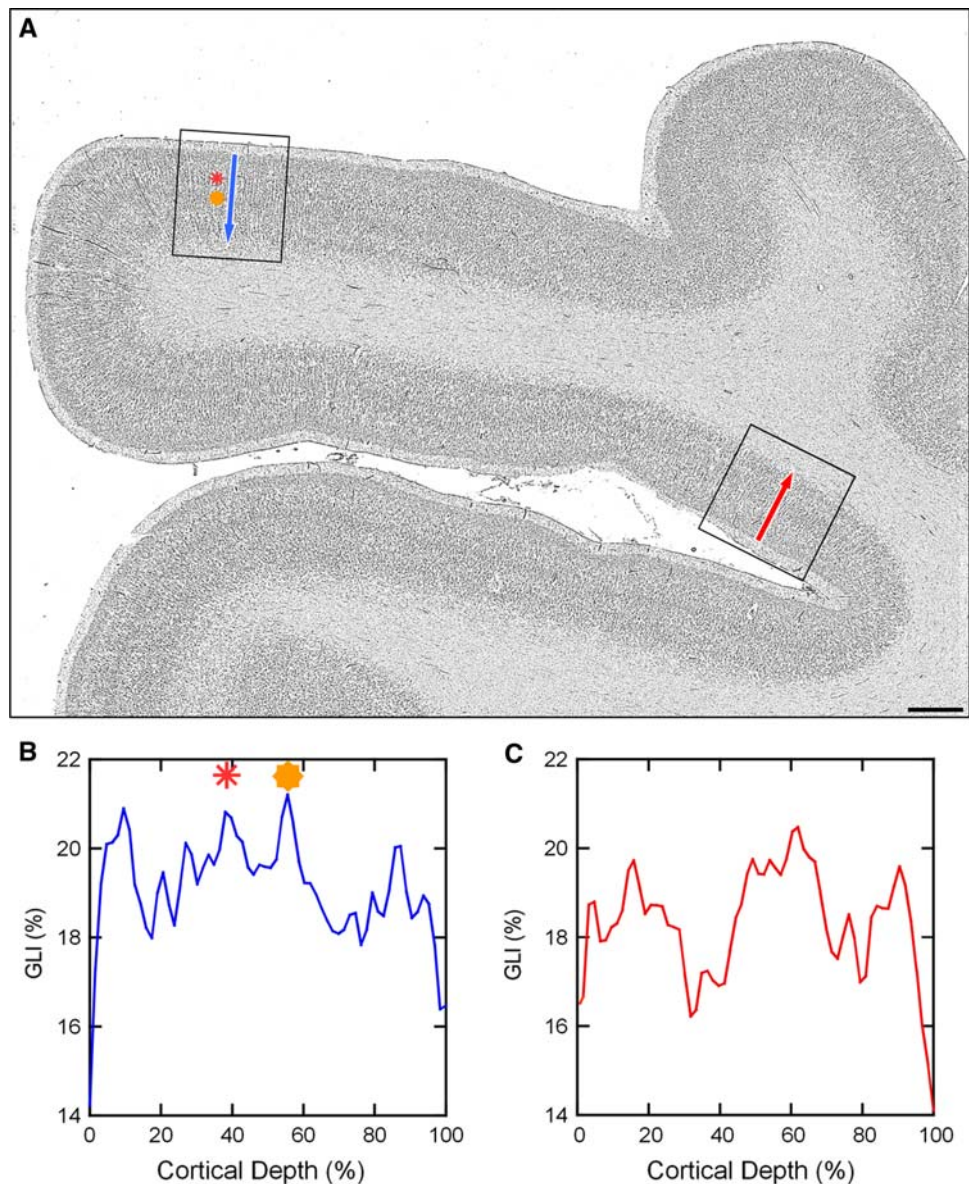
Two *GLI*-profiles sampled from two different cortical locations always differ in shape. Small, random differences may be caused by unspecific effects e.g. by local inhomogeneities in cell density not related to the layering pattern, interactions with vertical structures (columns, blood vessels) or artifacts in the histological specimen. Larger, systematic differences in shape are expected when comparing profiles sampled from two distinct cortical areas with distinct laminar patterns. To provide statistical evidence for shape dissimilarities specific to differences between areas, *GLI*-profiles are compared using distance analysis. This is illustrated in Fig. 7.

Features and Distance Analysis

Figure 7 shows the cytoarchitecture of the two cortical sectors. The mean *GLI*-profiles (Fig. 7c) represent the two laminar patterns. The peak in profile of cortical sector a) at an approximate cortical depth of 38% represents the main difference. It reflects the cell dense minicolumns in deep layer III which are missing in the profile of the second sector b).

Quantitative shape comparison is based on feature vectors with 10 elements extracted from each profile.

Fig. 5 *GLI*-profiles. **a** *GLI*-image of the histological section shown in Fig. 2. The image is inverted to improve visualization. Two cortical sectors are marked by black frames. The upper left arrow marks a cortical sector of the temporal plane, the lower right arrow a sector in the depth of the superior temporal sulcus. **b** Mean *GLI*-profile of the first cortical sector, calculated by averaging 20 *GLI*-profiles sampled from within this sector. Cell dense minicolumns in deep layer III and cell dense layer IV are marked by star symbols in **a** and **b**. The cortical depth (abscissa) is normalized to 100%. **c** Mean *GLI*-profile of the second cortical sector. Scale bar in **a** 1 mm



Feature vectors reduce the large number of points in a profile (i.e. 101) to a smaller number of features while preserving major information on the shape of the profile. We apply features based on statistical moments such as mean, standard deviation, skewness and kurtosis, calculated from the profile and the absolute of its first differential quotient (Schleicher et al. 1999). The two mean feature vectors (Fig. 7d) calculated from the two sets of feature vectors of the individual profiles quantify the laminar pattern in the two cortical sectors. The highest difference between pairs of corresponding features is found for the 9th feature (skewness of the differential quotient of the profile, 93% difference). It indicates the lamination of sector a) to be shifted towards the more superficial layers as compared to sector b) (Zilles et al. 2002a). This can also be recognized by comparing the two mean profiles in Fig. 7c.

For details of feature vectors calculated from moments and their interpretation see Zilles et al. (2002b). From the two sets of feature vectors, a distance measure is calculated. We apply the Mahalanobis distance (Bartels 1979). The Mahalanobis distance is a quantitative measure of dissimilarity of the laminar patterns in two cortical sectors which not only considers the mean shape, but also the variability in shape of the individual profiles in each sector. The higher the distance value, the higher the degree of dissimilarity.

Sliding Window and Distance Function

In a next step, the cortical ribbon is systematically covered by a set of traverses, 100 μm apart. Each traverse defines the course of a *GLI*-profile with starting and end points defined by two interactively traced contour lines (Fig. 8).



Fig. 6 Cytoarchitecture of the secondary auditory cortex, Brodmann's area 42. The *GLI*-profile of this cortical sector is shown in Fig. 5b. The layering pattern is characterized by pronounced minicolumns with high cell density in deep layer III and a high density of small granular cells in layer IV. Roman numerals indicate cortical layers. Scale bar 250 μ m

GLI-profiles are then extracted along the traverses using an in-house software written in MatLab for Windows (Version 7.0; Mathworks, USA).

The calculation of traverses considers the following constraints. They are not allowed to cross, and should run perpendicularly to each layer. They should also approach the course of the minicolumns in order to reduce noise in the profile introduced by interference between profiles and

minicolumns when a traverse crosses several columns. This is achieved by generating not linear (Schleicher et al. 2000), but curvilinear traverses using a differential equation from mathematical physics (Jones et al. 2000; Schmitt and Böhme 2002; Annese et al. 2004; Schleicher et al. 2005), as demonstrated in Fig. 8.

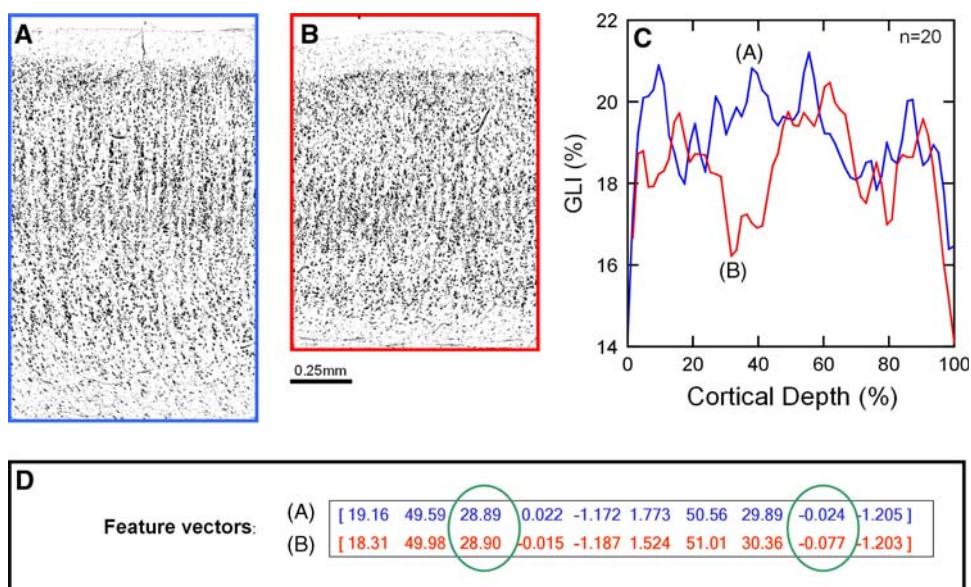
The cortical ribbon is now analyzed in a sliding window procedure using the *GLI*-profiles extracted. The sliding window is composed of two abutting cortical sectors each of which comprises b neighboring profiles (Fig. 8a; blocksize $b = 20$). Moving the sliding window across the cortical ribbon in one interprofile increments, Mahalanobis distance values are calculated at each position. The values are added to a Mahalanobis distance function, which describes the dependency of the distance on the position of the sliding window (Fig. 8b). Areal borders are expected at those positions where the sliding window is centered over an areal border and are indicated by local maxima in the distance function. They are tested for significance using a Hotelling's T^2 -test corrected for multiple comparisons (Schleicher et al. 1999).

The sliding window procedure is repeated with b increasing in a predefined range (e.g. $8 \leq b \leq 24$) in one profile increments. The lower limit 8 is set near the smallest number of profiles statistically required for group comparisons (10 features in two blocks of profiles); 24 limits the spatial resolution of the procedure to cortical areas not smaller than about the cortical thickness. This avoids interference with smaller, vertically oriented structures such as mini- and macrocolumns (Mountcastle 1978; Lübke and Feldmeyer 2007; Buxhoeveden and Casanova 2002).

Positions of putative areal borders are only considered if consistent maxima of the distance function with respect to

Fig. 7 Comparison of the laminar patterns shown in Fig. 5 using the consistent labels (A) and (B). **a** and **b**:

Cytoarchitecture of both sectors in identical magnification (scalebar 0.25 mm). **c** The two mean *GLI*-profiles are superimposed. **d** Two mean feature vectors calculated from the two sets of profiles ($n = 20$ profiles each). Each mean feature vector quantifies the laminar pattern of the corresponding cortical sector. Two pairs of features with similar values (left) and highly differing values (right) are marked. The Mahalanobis distance between the two sets of feature vectors is 3.41



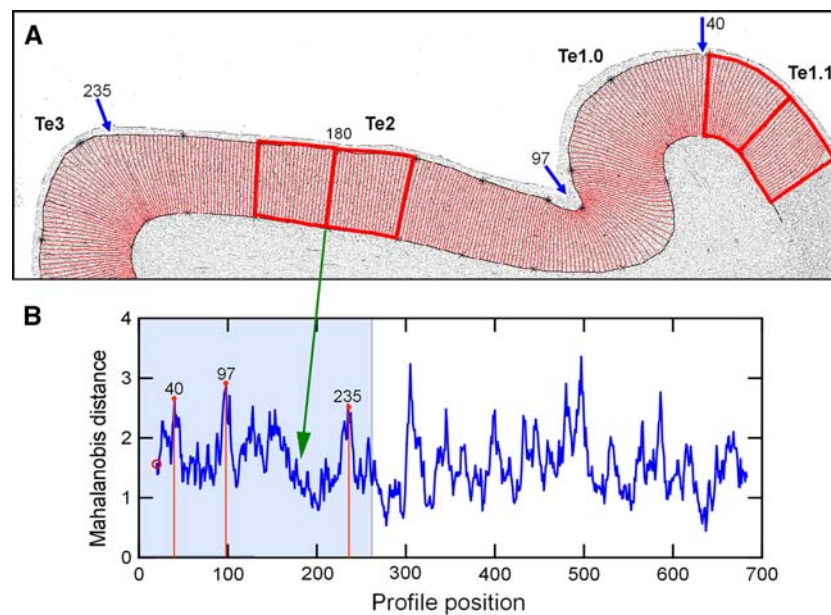


Fig. 8 Sliding window procedure. **a** Region of interest sampled by curvilinear traverses and *GLI*-profiles. The sliding window is comprised of two abutting cortical sectors marked by *outlines* with 20 profiles each. Two positions are shown: the starting position (*right*) and an intermediate position centered at profile 180. The *three short arrows* indicate the areal borders detected using the Mahalanobis distance function shown in **b** at profile positions 40, 97 and 235. Labels Te1.1, Te1.0, Te2 and Te3 denote the corresponding cortical

areas. For details see text. **b** Mahalanobis distance function generated by moving the sliding window across the cortical ribbon. A *long arrow* indicates the distance value at position 180. Statistically significant maxima are marked by *drop lines* and labeled with the corresponding position in the cortical ribbon shown in **a**. Only the shaded part of the distance function corresponds to the cortical ribbon shown in **a**, the remaining part results from the analysis of the cortex in the superior temporal sulcus (data not shown)

the size of the sliding window *b* were found (Amunts et al. 2002; Caspers et al. 2006). Three significant maxima indicating pronounced changes in laminar structure are detected in the example at positions 40, 97 and 235. The position of each maximum is plotted at the corresponding location in the cortical ribbon (Fig. 8a).

Such procedure revealed more areas in the human auditory region than shown in Brodmann's map. They were named according to a novel nomenclature (Morosan et al. 2005). Areas Te1.0 and Te1.1 correspond to BA 41, the putative structural correlate of the human primary auditory cortex. Te2 corresponds to BA 42. It is located on the dorsal surface of the superior temporal gyrus. Te3 is part of BA 22, but does not coincide with it. Unlike BA 22, Te3 does not extend onto the middle temporal gyrus.

Validating the Parcellation

The resulting parcellation pattern of the human superior temporal gyrus and TP is based on cytoarchitecture and does not include any *a priori* knowledge concerning the identification of the areas found. For a detailed description and interpretation of the areal parcellation established see Morosan et al. (2005).

We apply three strategies to validate the areal pattern found in an individual histological section: (1) visual, microscopical

inspection; (2) canonical analysis, and (3) consistency of areal borders across as set of consecutive sections.

- 1) Maxima of the Mahalanobis distance functions should closely match characteristic changes in cytoarchitecture between cortical areas as revealed by visual, microscopical inspection of the histological brain sections. The quantitatively defined border between Te2 and its laterally neighboring area Te3, for example, is found exactly at the position where the densely packed large pyramidal cells found in the midcortical layer IIIc of Te3 stopped (Figs. 9, 10). The comparison with other architectural modalities, e.g. with myelo- and receptorarchitectonic analyses of receptor binding sites of different neurotransmitters is an additional, independent way of validation. It results in a multimodal definition of a cortical area and has been described previously (Zilles et al. 2002a; Toga et al. 2006; see discussion).
- 2) A canonical analysis (Burwell 2001) of the examined areas enables a visualization the dissimilarities of the corresponding laminar patterns. Each data point in a scatterplot represents the feature vector of one *GLI*-profile from a distinct area. The canonical analysis constructs new, canonical variables and axes in a 2D scatterplot which are chosen so as to optimize the

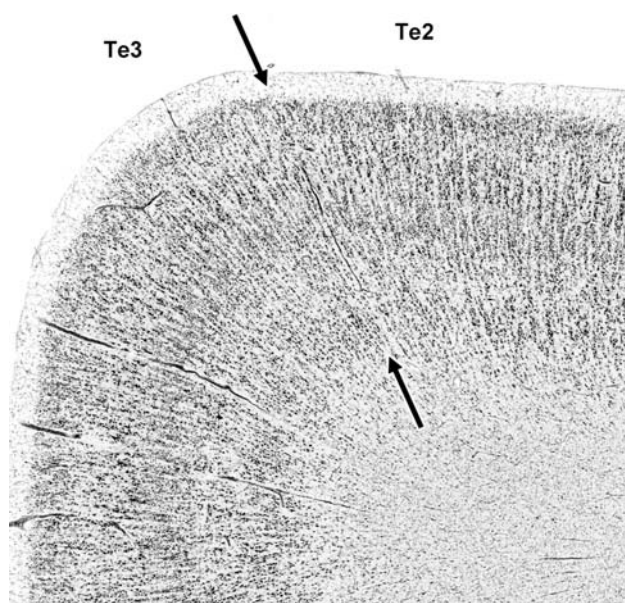


Fig. 9 Cytoarchitecture of the transition from the temporal plane (*right*) to the crown of the superior temporal gyrus (*left*). The arrows indicate the border between areas Te2 and Te3 detected by the sliding window procedure at profile position 235 shown in Fig 8

visualization of clusters. Distinct cortical areas exhibit

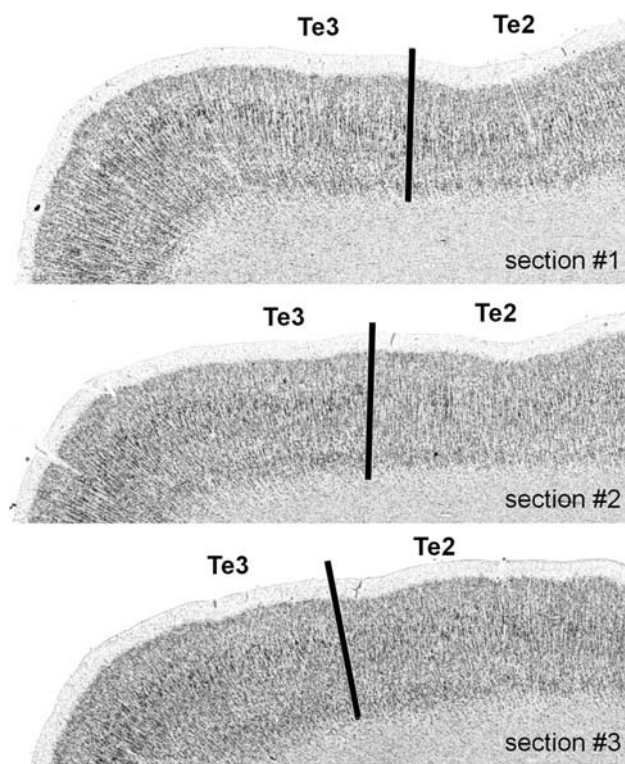


Fig. 10 Consistency of the border between areas Te2 and Te3 in a set of three neighboring sections. From section 1 to section 3, a shift to a more lateral position is obvious. 3D-reconstruction (data not shown) revealed this shift to describe the natural trace of this border on the brain surface

distinct clusters of *GLI*-profiles. This procedure is also helpful in determining a subset of shape describing features which contribute most to the separation between clusters. This subset indicates which cytoarchitectural properties show the highest differences between areas. A more detailed analysis of a canonical analysis for the validation of a parcellation of the human parietal operculum into four subdivisions in a sample of 10 brains is given in Eickhoff et al. (2006).

- 3) Putative areal borders are checked for consistency in a set of consecutive sections (e.g. Geyer et al. 1999; Malikovic et al. 2007). This excludes false positive borders, since the probability of detecting random borders at comparable locations in a set of sections is exceedingly small. Fig. 10 compares the position of the Te2/Te3 border across three consecutive sections. A slight shift to a more lateral position in the 3rd section can be seen in comparison to the first section. 3D-reconstructions (data not shown) revealed this shift to describe the natural trace of this border on the brain surface.

3D Maximum Probability Maps and Brain Atlas

The areal patterns of areas Te1, Te2 and Te3 in a sample of 10 brains were compiled by a maximum probability map procedure (Eickhoff et al. 2005). To that end, the individual brains and their areal patterns were warped to a common 3D reference brain (Collins et al. 1994; Holmes et al. 1998). Figure 11 shows areas Te1, Te2 and Te3 in representative coronal section of this reference brain. Only Te3 is visible on the outer surface, Te1 and Te2 are hidden

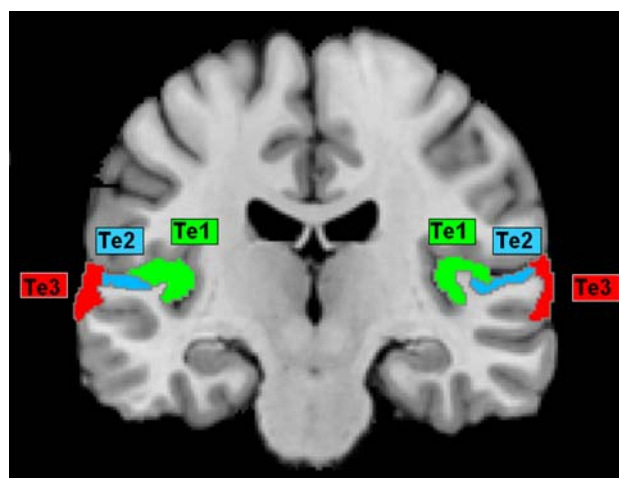


Fig. 11 Maximum probability map of areas Te1, Te2 and Te3 in a sample of 10 brains, depicted in a representative coronal section of the reference brain. Sides as shown in the graph. For details see text

in the lateral sulcus. In a maximum probability map, each voxel is assigned to the area with the maximum frequency in this voxel. An example: a given voxel in the reference brain represents Te1 in one, Te2 in 7 and Te3 in two brains. Therefore, this voxel is assigned to area Te2. The mapping of the human auditory cortices (Morosan et al. 2005) was performed within the framework of the Düsseldorf/Jülich brain atlas project (Amunts and Zilles 2006). For details of this novel probabilistic atlas see Amunts et al. (2007). Data and maps are available for download from www.fz-juelich.de/inm/index.php?index=51.

Discussion

In the past five decades, stereology has added valuable tools to quantitative, cytoarchitectural analysis of cortical tissue. Two main groups of applications can be distinguished. A first group comprises highly sophisticated procedures based on elaborated image analyzing approaches used to examine various aspects of distinct cortical entities. A typical example is the analysis of minicolumns in normal and pathological brains. A variety of parameters of the spatial, minicolumnar conformation such as the columnar width and the intercolumnar distance were analyzed in a study of Buxhoeveden et al. (2000). Casanova et al. (2007) investigated the intercellular distance (median free path) within minicolumns throughout development and aging. A study of Casanova et al. (2006) reported evidence of a deviation of minicolumnar structure in autistic brains from normal brains. These approaches are specific to distinct cortical entities and have to be adjusted to the pronounced anisotropy of most cortical structure in order to avoid biased estimates. An example: the cutting of histological sections parallel to the minicolumnar orientation is mandatory for an unbiased estimate of most minicolumnar properties. This constraint was tempered by Casanova and Switala (2005) who published a procedure to estimate the deviation of the plane of sectioning from the minicolumnar orientation. By calculating the amount of minicolumnar fragmentation caused by oblique cutting of minicolumns, inadequate sections could be detected and discarded from the analysis.

A second group of procedures aims at robust, generalized applications which are not affected by anisotropy and can therefore analyze standard histological serial sections without preferential orientation to the cortical surface. A typical example is the *gray level index (GLI)*, an estimate of the local volume density of neurons. The following citations refer to some typical applications using the *GLI*.

Characterization and quantitative cytoarchitecture of cortical areas in normal brains of various species were described in Schleicher et al. (1978), Amunts et al. (1999),

Sherwood et al. (2004) and Schleicher and Zilles (2005). A first step towards algorithm-based brain mapping was the enhanced visualization of the laminar pattern in various species by Rehkämper et al. (1984), Sauer (1983a, b), Zilles et al. (1978, 1980).

Changes in cytoarchitecture induced by aging and pathological processes were quantified in the studies of Kawasaki et al. (1997), Kawasaki et al. (2000), Casanova et al. (2002), Vogeley et al. (2003) and by Casanova et al. (2006).

Studies on interhemispheric asymmetry and handedness based on *GLI* analysis were published by Amunts et al. (1996) and Amunts et al. (1997). *GLI*-analysis proved to be sensitive to developmental changes and disorders in a study of Amunts et al. (1997). Comparisons of cortical cytoarchitecture across species were published in Armstrong et al. (1986) and Zilles et al. (1982).

The presently described algorithm-based mapping approach based on *GLI*-profiles and distance analysis has been successfully applied to more than 40 areas of the human isocortex. The reliability, consistency and precision of the established cytoarchitectonic borders was validated by microscopic, visual inspection, multivariate statistics and 3D-reconstruction. In addition, many cytoarchitectonic borders have been confirmed by independent receptor architectonic studies, where the areal patterns have been defined based on regional differences in the distribution of densities of receptor binding sites of various neurotransmitter systems (Geyer et al. 1996; Zilles et al. 2002b; Geyer and Zilles 2005; Palomero-Gallagher et al. 2008). Morosan et al. (2005) applied this technique to validate the areal pattern found on the superior temporal gyrus. Analyzing the spatial distribution of various transmitter receptors, main borders coincided with those found in cytoarchitecture, and robust subparcellations were added. The close agreement of the border between human auditory areas Te3 and Te4 in cytoarchitecture and receptorarchitecture is illustrated in Fig. 12. This finding suggests additional, functional validity to the cytoarchitectonic parcellation (Amunts et al. 2007).

The sliding window procedure is open to integrate a variety of image processing algorithms to exclude distinct image compartments from the analysis prior to extracting density profiles. In order to adjust the procedure to pathologies which are characterized by non-homogeneous increments of non-neuronal elements, size and shape analysis are optionally included to eliminate non-neuronal elements, provided the specimens enable the differentiation of neurons from glia and endothelial cells.

Algorithm-based mapping procedures are not limited to cytoarchitecture and receptorarchitecture, but have been applied to a variety of architectural modalities from which images can be acquired which reveal a quantifiable,

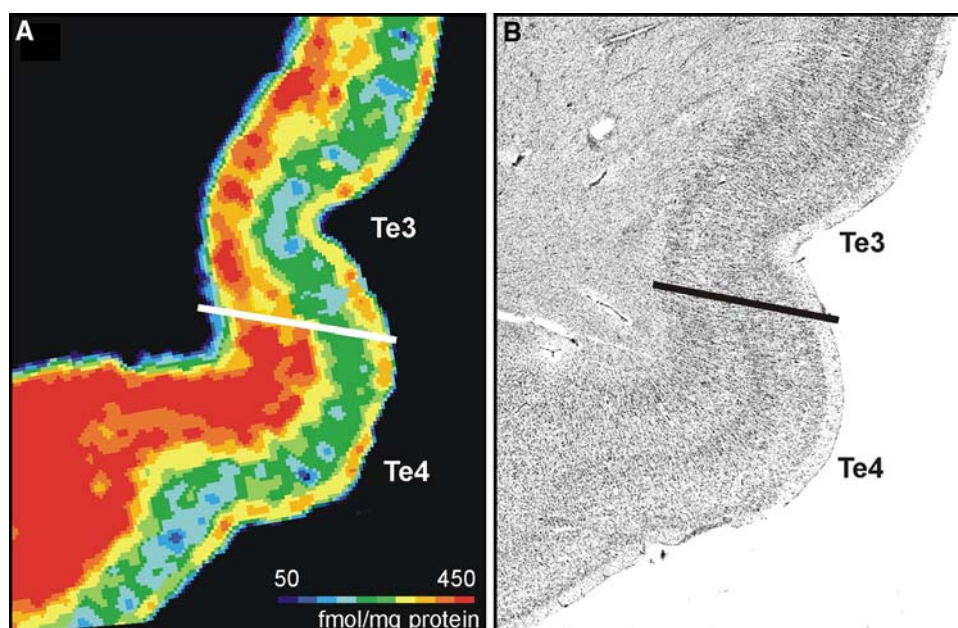


Fig. 12 Localization of the border between the higher auditory area Te3 and the multimodal area Te4 on the basis of receptor architecture (**a**) and cytoarchitecture (**b**). The areal border is characterized by a sharp increase in the concentration of noradrenergic $\alpha 1$ receptors from Te3 to Te4, which is particularly found in the infragranular layers **a**. The receptor was visualized using [^3H]-prazosin. The concentration of the total binding of this ligand in fmol/mg protein is color or gray

coded. The same border is also marked by a decrease in cell packing densities from Te3 to Te4, which is observed in the supragranular as well as in the infragranular layers **b**. Note that the position of the border between Te3 and Te4 is found at almost the same position, irrespective of the modality used to segregate the two areas from each other

laminar pattern. Examples are stainings for acetylcholinesterase (Hackett et al. 2001) and myeloarchitecture (Annese et al. 2004; Schmitt et al. 2004).

Recent progress in MRI-technology has improved the spatial resolution of in vivo MRI-scans down to less than 300 μm (Walters et al. 2007). This enabled the laminar pattern to become visible and analyzed in MR-images of the living human brain. An example of the presently described sliding window procedure for the identification of the motion sensitive area V5/MT in MR-images in four human hemispheres were published by Walters et al. (2007). The architectural finding showed a high spatial correlation with distinct activations found in functional imaging. In another study, results from a functional imaging study analyzing moving dot patterns (Wilms et al. 2005) were spatially correlated with the structural mapping of area hoc5 (Malikovic et al. 2007).

These applications have successfully introduced algorithm-based, architectural mapping procedures. According to our experience in analyzing large samples in many brains and species, developments and improvements with respect to the following two issues should be considered.

Cortical folding affects the laminar structure with respect to layer widths and cell densities (Bok and van Kip 1939; de Vos et al. 2004). The mapping procedure

described presently is not corrected for these conformational changes. Simulation studies in a model cortex, however, revealed the sliding window procedure to be sensitive to sharp areal borders and, to a certain extent, robust to smooth changes in laminar pattern caused by cortical folding (Schleicher et al. 1999). In this study, the Mahalanobis distance analysis was superior to other distance measures such as the Euclidean distance with respect to its robustness in folded cortices. Areal borders are also detected in unfolded parts of the cortical ribbon. The complete mapping study of the human superior temporal gyrus (Morosan et al. 2005) gives some examples. Borders detected at the crown of a gyrus (Fig. 9) or in the depth of a sulcus must be carefully validated. The cytoarchitectural border depicted in Fig. 9 was validated by distinct changes in the laminar pattern revealed at microscopic inspection. If false positive borders occur, they are preferably located at sites with a high degree of curvature. Corrections of layer widths and cell densities for folding effects would increase the sensitivity of mapping procedures and reduce such false positive border detections.

To date, structural mapping procedures analyze 2D images from 3D brains. This reduction in dimensionality results in a specific problems: in serial sections, border detection is hampered by distortions of the laminar pattern

caused by the unavoidable oblique or tangential cutting of the cortical ribbon. Severe distortions can be recognized by visual inspection and these ROIs can then be excluded from the analysis. Thus, serial sections of gyrified brains cannot be completely mapped using a single plane of sectioning. Therefore, the extension of mapping procedures to 3D is highly desirable. The generation of 3D, curvilinear traverses in MRI-volumes was published by Jones et al. (2000) and can also be applied to histological volumes. However, the reconstruction of such histological volumes from serial sections appropriate for algorithm-based border detection requires image registration techniques which must be precise enough to match individual cell positions across sections. These techniques were not available until now, but recent progress in imaging techniques and registration software (Pitiot et al. 2006) makes this a realistic target.

Conclusion

Quantitative cytoarchitecture based of profile analysis offers a variety of applications beyond the scope of cortical parcellation. Once the areal pattern is established, profile analysis enables the comparison of lamination pattern between distinct cortical areas. Given that cortical areas are defined by their unique lamination pattern and that borders between architectonic areas are functionally relevant (Luppino et al. 1991; Nelissen et al. 2005), the analysis of similarities and dissimilarities between the lamination pattern of cortical areal opens a new dimension for functional, comparative, developmental and pathological studies on the human neocortex and helps to understand the cortical organization more deeply. Amunts et al. (2007) sampled *GLI*-profiles extracted from BA 17 and BA 18 (Amunts et al. 2000), areas hOc3v and hOc4v from the extrastriate cortex (Kujovic, personal communication) and area hOc5, the structural correlate of the complex V5/MT (Malikovic et al. 2007), and subjected them to a hierarchical cluster analysis. The clustering of the five areas reflected the segregation of the visual cortex into a dorsal and ventral stream, which is correlated to connectivity and function (Ungerleider and Mishkin 1982). This example demonstrated how quantitative cytoarchitectural analysis reflects the inner organization of cortical areas.

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